

Qualification Pack



Electrician (Mineral Sands)

QP Code: MIN/Q3104

Version: 1.0

NSQF Level: 4.5

SCMS (Skill council of Mining Sector) || FIMI House, B-311, Okhla Industrial Area, Phase-I
New Delhi-110020 || email:scms@skillcms.in

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MIN/Q3104: Electrician (Mineral Sands)

Brief Job Description

An Electrician (Mineral Sands) is responsible for the Operation and Maintenance (O&M) of the installed equipment and electrical supply system.

Personal Attributes

The individual should be physically fit to work for long durations. The person should have attention to detail, organizational and problem-solving skills. The individual should also have appropriate verbal and written communication skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N3107: Test and operate electrical equipment and motors at the dredge and supplementary mining plant](#)
2. [MIN/N3108: Test and operate electrical equipment and motors at the mineral separation plant](#)
3. [MIN/N3109: Test and operate electrical equipment and motors at the rare earth extraction plant](#)
4. [MIN/N3110: Test and operate electrical equipment and motors at the main receiving sub-station](#)
5. [MIN/N1810: Follow the applicable health, safety and environmental guidelines \(Placer Mining\)](#)
6. [MIN/N1811: Work collaboratively and effectively in a team](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
Country	India
NSQF Level	4.5



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Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 7411.0200
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) ((in Electrical)) OR Completed 3 year diploma after 10th ((in Electrical)) OR 10th grade pass (with 3 year of any combination of NTC/NAC/CITS or equivalent (in Electrical))
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	14/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-4.5-MG-02175-2024-V1-SCMS
NQR Version	1.0

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MIN/N3107: Test and operate electrical equipment and motors at the dredge and supplementary mining plant

Description

This occupational standards unit is about testing and operating electrical equipment and motors at the dredge and supplementary mining plant.

Scope

The scope covers the following :

- Operate the processing equipment
- Test cables, joints and switchgears
- Perform equipment maintenance

Elements and Performance Criteria

Operate the processing equipment

To be competent, the user/individual on the job must be able to:

- PC1.** monitor the operation of dredge and different types of slurry pumps
- PC2.** check the operation of different types of 3-phase induction motors at the dredge
- PC3.** test and maintain the 3-phase Induction motors engaged in the winching system
- PC4.** carry out overhauling of the motor including greasing, bearing replacement, etc
- PC5.** use the appropriate electrical tools and measuring instruments for the installation of electrical systems and equipment
- PC6.** install electrical panels as per the relevant drawings
- PC7.** check and ensure the correct operation of the newly installed equipment
- PC8.** coordinate with the Main Receiving Sub-Station (MRSS) regarding power supply
- PC9.** follow the appropriate safety measures to protect the electrical systems during extreme weather and natural calamities

Test cables, joints and switchgears

To be competent, the user/individual on the job must be able to:

- PC10.** check the condition of Cross-linked Polyethylene (XLPE) , and Polyvinyl Chloride (PVC) insulated cables
- PC11.** install and check the condition of different types of trailing cables, e.g. 11kV, 3.3kV, etc.
- PC12.** connect cables to panels and sub-panels appropriately
- PC13.** carry out cable end termination in control panels appropriately
- PC14.** install different types of cable joints and cable trays
- PC15.** inspect, identify and rectify abnormal conditions in the circuits
- PC16.** check insulation resistance for different types of installations, e.g. control panel, motor and cables
- PC17.** check for the correct functioning of the toggle switch, push button switch, selector switch, cam switch, etc.

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PC18. install different types of circuit breakers, such as Miniature Circuit Breaker (MCB), Molded Case Circuit Breakers (MCCB), Earth-leakage Circuit Breaker (ELCB), Air Circuit Breaker (ACB), SF6 circuit breaker, and Vacuum Circuit Breaker (VCB)

Perform equipment maintenance

To be competent, the user/individual on the job must be able to:

- PC19.** carry out testing and maintenance of various ratings of transformers, ensuring their protection
- PC20.** carry out the maintenance of the lightening arrestor, insulated conductor, disk insulator, pin insulator, cross arm, etc
- PC21.** carry out the maintenance of Overhead (OH) lines and their protections
- PC22.** carry out the maintenance of the earthing system
- PC23.** measure the earth pit resistance using the appropriate tool and following the standard procedure
- PC24.** carry out the maintenance and protection of dredge main motor: 3.3kW, 315kW, 525kW, 415V
- PC25.** carry out the maintenance of 3.3kV and 11kV switch gears
- PC26.** carry out the maintenance and operation of different types of relays, and Variable Frequency Drive (VFD)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the working principle, testing, operation, maintenance and protection of different types of 3-phase induction motors (e.g. 3-phase induction motors engaged in winching system) at the dredge
- KU2.** the use of a mine organization chart
- KU3.** the importance of discipline and punctuality in mine operations
- KU4.** the health and safety standards applicable to mine operations, e.g. the appropriate safety measures to be taken in the vicinity of working machinery, and the use of relevant PPE
- KU5.** the basic concepts regarding AC single-phase and 3-phase supply
- KU6.** the conversion of the star to delta and vice versa
- KU7.** Direct on line (DOL) and forward, and reverse motors connections
- KU8.** different types of electric lighting and heating systems for commercial, domestic residential and industrial use
- KU9.** how to measure current, voltage, resistance, impedance, power factor and energy
- KU10.** the classification and identification of the electrical equipment, cables, wires and electrical accessories used in industry
- KU11.** the organizational procedure for requisitioning and indenting materials and equipment
- KU12.** the appropriate personnel to contact for queries concerning procurement, store management, quality management, and inventory management
- KU13.** different electrical units or specifications, such as wattage, resistance, voltage, frequency, current, Amp Hours (Ah), Watt Hours (Wh), etc
- KU14.** the use of circuit diagrams and blueprints for the installation of electrical systems

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- KU15.** different types of tools required for dredging and mining operations
- KU16.** their applications of various types of motors, e.g. AC, DC, and special motors
- KU17.** different types of transformers, cooling of transformers, dielectric capacity
- KU18.** breakers and their troubleshooting
- KU19.** how to determine the fuse size according to the load of the circuit and its location
- KU20.** the classification, insulation and the types of cables
- KU21.** the types of faults and location of faults in cables
- KU22.** different types of circuit breakers and causes of faults in panels
- KU23.** the types and methods of earthing
- KU24.** how to use megger and test lamps for locating faults

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the critical equipment observations and report them to the supervisor
- GS2.** write requisition to supervisor on the requirement of apparatus, tools etc
- GS3.** read and interpret blueprint, control diagrams and SOP documents
- GS4.** communicate politely and professionally
- GS5.** listen attentively to understand the information/ instructions being shared
- GS6.** plan and prioritize tasks to ensure their timely completion
- GS7.** systematize all process/equipment in 5s plan to process fast
- GS8.** identify potential work-related problems and take appropriate preventive measures
- GS9.** evaluate all possible solutions to a problem and implement the best one
- GS10.** make appropriate decisions concerning work within the applicable limits of authority
- GS11.** follow the organizational procedure during workplace emergencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate the processing equipment</i>	10	15	-	6
PC1. monitor the operation of dredge and different types of slurry pumps	1	1	-	-
PC2. check the operation of different types of 3-phase induction motors at the dredge	1	2	-	1
PC3. test and maintain the 3-phase Induction motors engaged in the winching system	2	2	-	1
PC4. carry out overhauling of the motor including greasing, bearing replacement, etc	1	1	-	1
PC5. use the appropriate electrical tools and measuring instruments for the installation of electrical systems and equipment	1	1	-	-
PC6. install electrical panels as per the relevant drawings	1	3	-	1
PC7. check and ensure the correct operation of the newly installed equipment	1	1	-	1
PC8. coordinate with the Main Receiving Sub-Station (MRSS) regarding power supply	1	2	-	1
PC9. follow the appropriate safety measures to protect the electrical systems during extreme weather and natural calamities	1	2	-	-
<i>Test cables, joints and switchgears</i>	10	20	-	8
PC10. check the condition of Cross-linked Polyethylene (XLPE) , and Polyvinyl Chloride (PVC) insulated cables	1	2	-	1
PC11. install and check the condition of different types of trailing cables, e.g. 11kV, 3.3kV, etc.	1	3	-	1
PC12. connect cables to panels and sub-panels appropriately	1	2	-	-
PC13. carry out cable end termination in control panels appropriately	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. install different types of cable joints and cable trays	1	2	-	1
PC15. inspect, identify and rectify abnormal conditions in the circuits	1	2	-	1
PC16. check insulation resistance for different types of installations, e.g. control panel, motor and cables	1	2	-	1
PC17. check for the correct functioning of the toggle switch, push button switch, selector switch, cam switch, etc.	1	3	-	1
PC18. install different types of circuit breakers, such as Miniature Circuit Breaker (MCB), Molded Case Circuit Breakers (MCCB), Earth-leakage Circuit Breaker (ELCB), Air Circuit Breaker (ACB), SF6 circuit breaker, and Vacuum Circuit Breaker (VCB)	2	3	-	2
<i>Perform equipment maintenance</i>	10	15	-	6
PC19. carry out testing and maintenance of various ratings of transformers, ensuring their protection	1	1	-	-
PC20. carry out the maintenance of the lightening arrester, insulated conductor, disk insulator, pin insulator, cross arm, etc	2	3	-	1
PC21. carry out the maintenance of Overhead (OH) lines and their protections	1	1	-	1
PC22. carry out the maintenance of the earthing system	1	1	-	-
PC23. measure the earth pit resistance using the appropriate tool and following the standard procedure	1	2	-	1
PC24. carry out the maintenance and protection of dredge main motor: 3.3kW, 315kW, 525kW, 415V	1	3	-	1
PC25. carry out the maintenance of 3.3kV and 11kV switch gears	1	2	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. carry out the maintenance and operation of different types of relays, and Variable Frequency Drive (VFD)	2	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3107
NOS Name	Test and operate electrical equipment and motors at the dredge and supplementary mining plant
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N3108: Test and operate electrical equipment and motors at the mineral separation plant

Description

This occupational standards unit is about testing and operating electrical equipment and motors at the mineral separation plant

Scope

The scope covers the following :

- Supply electricity to the installed system
- Perform maintenance and operations

Elements and Performance Criteria

Supply electricity to the installed system

To be competent, the user/individual on the job must be able to:

- PC1.** check the electrical layout and single-line circuit diagram
- PC2.** check for the correct installation of the High-Tension Separator, Electrostatic Plate Separator, Magnetic Separator, Induced Roll Magnetic Separator, Rare Earth Drum Magnetic Separator, Cross Belt Magnetic Separator, High-Intensity Induced Roll Magnetic Separator, etc.
- PC3.** follow the supervisor's work instructions/ work-related documents to determine the work requirements and work accordingly
- PC4.** measure current, voltage and resistance of single-phase and three-phase system
- PC5.** identify the phase, neutral and earth connections in the supply system
- PC6.** select and maintain ISO Standard hand tools and measuring tools required at the site, e.g., insulation resistance tester or megger, clamp meter, multimeter, voltage probe, etc
- PC7.** ensure power system protection for induction motors, transformers, etc

Perform maintenance and operations

To be competent, the user/individual on the job must be able to:

- PC8.** overhaul the motor
- PC9.** grease and replace the bearing
- PC10.** maintain the UPS, battery bank and capacitor bank in an appropriate condition
- PC11.** conduct different types of tests on motors, e.g. IR Value, No load, Winding resistance, Polarity Test, Blocked Rotor Test etc
- PC12.** check cable laying, cable jointing and terminations, and carry out their maintenance
- PC13.** carry out the maintenance of different types of breakers, panels etc
- PC14.** check the rotation of the driving system in slurry pumps and booster pumps
- PC15.** calibrate the ammeter, kilo-volt meter, etc
- PC16.** conduct the insulation test
- PC17.** check the transformer oil level and replenish it as required

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- PC18.** install different types of light fittings for illumination, e.g. fluorescent light, metal halide, high-pressure mercury vapour, high-pressure sodium vapour, LED, flameproof lighting fixtures in hazardous areas
- PC19.** lay different types of cables, selecting them based on their specifications and current carrying capacity
- PC20.** verify the characteristics of series and parallel circuits
- PC21.** use direct on-line starter, motor forward reverse, star-delta starter, and soft starter, as appropriate
- PC22.** carry out maintenance and programming of Variable Frequency Drive (VFD)
- PC23.** use electrical supply systems and components including transformers, generators, circuit breakers, isolators, bus bars, wiring, fuses, switchboards, control panels, relays etc. as per the required specifications
- PC24.** use the appropriate tools to measure voltage, current, power, energy, frequency, RPM, etc
- PC25.** obtain a work permit from the competent/ authorized person, following the applicable procedure
- PC26.** check the insulation resistance for different types of installations, e.g. control panel, motor wiring, etc.
- PC27.** plan and ensure proper ducting and wiring as per the Indian Electricity Rules (IER)
- PC28.** carry out maintenance and identify faults in the electrical circuits, e.g. High Tension Separator, Electrostatic Plate Separator, Magnetic Separator, Induced Roll Magnetic Separator, Rare Earth Drum Magnetic Separator, Cross Belt Magnetic Separator, High-Intensity induced Roll Magnetic Separator, etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the types of motors and their insulation
- KU2.** different types of connections (Star& delta)
- KU3.** the types of enclosure, and their duty cycle
- KU4.** different types of cables, their specifications and current carrying capacity
- KU5.** the relevant health and safety requirements applicable in the workplace
- KU6.** the escalation matrix and procedures for reporting work-related issues
- KU7.** different types of switches, such as SPST, SPDT, DPST, DPDT, Toggle switch, push button switch, level Actuator, limit switch, selector switch, flame-proof switch, etc
- KU8.** the types and applications of circuit breakers, such as Miniature Circuit Breaker (MCB), Molded, Case Circuit Breakers (MCCB), Earth-leakage Circuit Breaker (ELCB), Air Circuit Breaker (ACB), SF6 circuit breaker, Vacuum Circuit Breaker (VCB), flame-proof switchgear
- KU9.** the types, sizes, current rating, other specifications and applications of different types of cables
- KU10.** different types of joints in cables, such as straight through, long routes and repaired sections, Branch y joints for branches of a sector, t-joints for branching of a sector, transition joints, and special joints between two different types of cables
- KU11.** how to plan the installation work using drawings and other relevant documentation

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- KU12.** the types and joints of conductors and insulators
- KU13.** wire gauge, conductor material and its current carrying capacity
- KU14.** how to determine the fuse size, MCB, MCCB etc., according to the load of circuits and their location
- KU15.** the use of Current Transformer (CT), Potential Transformer (PT), fuse base and link, connectors, indicator lights and push buttons
- KU16.** the process of isolating different electrical equipment
- KU17.** the significance of safe tripping/ interlocking mechanism in preventing electrical fires
- KU18.** the importance of providing double earthing for 3-phase equipment
- KU19.** the importance and process of installing cables and equipment as per IER-1910, IEA1956, and NEC1985 & ESA1948

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make note of work-related observations
- GS2.** interpret the relevant drawings, sketches, symbols, measurements and Original Equipment Manufacturer (OEM) manuals
- GS3.** use the online Enterprise Resource Planning (ERP) systems under guidance of the supervisor
- GS4.** communicate politely and professionally
- GS5.** use effective questioning techniques to extract maximum information about different tasks
- GS6.** plan daily tasks appropriately to ensure their timely completion
- GS7.** undertake different activities as per the applicable task lists and schedules with minimum supervision
- GS8.** make appropriate decisions concerning the area of work
- GS9.** follow the 5s principles at work
- GS10.** use reasoning skills to identify and resolve basic problems at work
- GS11.** assist in identifying and implementing appropriate solutions to work-related problems
- GS12.** follow the organizational protocol during emergencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Supply electricity to the installed system</i>	8	15	-	5
PC1. check the electrical layout and single-line circuit diagram	1	3	-	1
PC2. check for the correct installation of the High-Tension Separator, Electrostatic Plate Separator, Magnetic Separator, Induced Roll Magnetic Separator, Rare Earth Drum Magnetic Separator, Cross Belt Magnetic Separator, High-Intensity Induced Roll Magnetic Separator, etc.	1	4	-	2
PC3. follow the supervisor's work instructions/ work-related documents to determine the work requirements and work accordingly	1	1	-	-
PC4. measure current, voltage and resistance of single-phase and three-phase system	1	2	-	-
PC5. identify the phase, neutral and earth connections in the supply system	1	2	-	1
PC6. select and maintain ISO Standard hand tools and measuring tools required at the site, e.g., insulation resistance tester or megger, clamp meter, multimeter, voltage probe, etc	2	2	-	1
PC7. ensure power system protection for induction motors, transformers, etc	1	1	-	-
<i>Perform maintenance and operations</i>	22	35	-	15
PC8. overhaul the motor	1	-	-	-
PC9. grease and replace the bearing	1	-	-	-
PC10. maintain the UPS, battery bank and capacitor bank in an appropriate condition	1	1	-	1
PC11. conduct different types of tests on motors, e.g. IR Value, No load, Winding resistance, Polarity Test, Blocked Rotor Test etc	1	2	-	1
PC12. check cable laying, cable jointing and terminations, and carry out their maintenance	1	2	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. carry out the maintenance of different types of breakers, panels etc	1	2	-	1
PC14. check the rotation of the driving system in slurry pumps and booster pumps	1	2	-	1
PC15. calibrate the ammeter, kilo-volt meter, etc	1	1	-	1
PC16. conduct the insulation test	1	3	-	1
PC17. check the transformer oil level and replenish it as required	1	2	-	-
PC18. install different types of light fittings for illumination, e.g. fluorescent light, metal halide, high-pressure mercury vapour, high-pressure sodium vapour, LED, flameproof lighting fixtures in hazardous areas	1	4	-	1
PC19. lay different types of cables, selecting them based on their specifications and current carrying capacity	1	3	-	1
PC20. verify the characteristics of series and parallel circuits	1	1	-	-
PC21. use direct on-line starter, motor forward reverse, star-delta starter, and soft starter, as appropriate	1	2	-	-
PC22. carry out maintenance and programming of Variable Frequency Drive (VFD)	1	2	-	1
PC23. use electrical supply systems and components including transformers, generators, circuit breakers, isolators, bus bars, wiring, fuses, switchboards, control panels, relays etc. as per the required specifications	1	2	-	1
PC24. use the appropriate tools to measure voltage, current, power, energy, frequency, RPM, etc	1	1	-	-
PC25. obtain a work permit from the competent/ authorized person, following the applicable procedure	1	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. check the insulation resistance for different types of installations, e.g. control panel, motor wiring, etc.	1	1	-	1
PC27. plan and ensure proper ducting and wiring as per the Indian Electricity Rules (IER)	1	2	-	-
PC28. carry out maintenance and identify faults in the electrical circuits, e.g. High Tension Separator, Electrostatic Plate Separator, Magnetic Separator, Induced Roll Magnetic Separator, Rare Earth Drum Magnetic Separator, Cross Belt Magnetic Separator, High-Intensity induced Roll Magnetic Separator, etc	2	2	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3108
NOS Name	Test and operate electrical equipment and motors at the mineral separation plant
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N3109: Test and operate electrical equipment and motors at the rare earth extraction plant

Description

This occupational standards unit is about testing and operating electrical equipment and motors at the rare earth extraction plant

Scope

The scope covers the following :

- Supply electricity to the installed system and equipment
- Perform testing, maintenance and operations

Elements and Performance Criteria

Supply electricity to the installed system and equipment

To be competent, the user/individual on the job must be able to:

- PC1.** use the appropriate Personal Protective Equipment (PPE) including Gum boots, Polyurethane (PU) gloves, ear muffs and eye protection
- PC2.** follow the appropriate electrical safety precautions at the rare earth extraction plant
- PC3.** obtain the work permit from the authorized person following the applicable procedure
- PC4.** maintain the electrical checkpoints
- PC5.** ensure the relevant electrical parameters are aligned with the process requirement of 3-phase induction motors at various locations in the rare earth extraction plant
- PC6.** check the circuit breakers, isolators, bus bars, wiring, fuses, earthing, switchboard, control panels, relays, and diesel generators for the correct functioning
- PC7.** follow the applicable safety standards in the earthing system, armoured cables and flexible cables as per the Indian Electricity Rules
- PC8.** maintain the electrical equipment, such as motors, fans, lighting, ACs, heaters, compressors, MCC Panels, LDB, PDB, etc
- PC9.** inspect the electrical equipment to identify electrical risks, hazards, defects and the needs for the adjustment/repair
- PC10.** modify the parameters of electrical equipment, if required to ensure alignment with the prescribed standards

Perform testing, maintenance and operations

To be competent, the user/individual on the job must be able to:

- PC11.** carry out maintenance of different types of motors and pumps, e.g. agitators motors, crystallize motors, centrifugal pumps, evaporator pumps, process water pumps, etc
- PC12.** carry out predictive, preventive and breakdown maintenance of diesel generators, motors, circuit breakers, isolators, bus bars, MCC panels, switchboards, wiring, protective relays, etc
- PC13.** carry out testing and identification of electrical systems for proper functioning of electrical wiring, equipment, and fixtures

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- PC14.** repair or replace defective wiring in power sources, lighting, AC and safety control systems, following the standard procedure
- PC15.** follow appropriate safety precautions to ensure personal protection from corrosive liquids and gases found in the rare earth extraction plant

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of chemical processes carried out in a chemical plant, e.g. Attacking Operation, Leaching Operation, Crystallization, Centrifuge, TSP Flash Dryer, Evaporation, Deactivation, RE Evaporation, Filtration etc
- KU2.** the importance of discipline and punctuality in mining operations
- KU3.** the importance of following the defined procedures / daily maintenance checklist
- KU4.** different types of cables, electrical enclosures, and accessories used in chemical plants
- KU5.** the appropriate safety precautions to ensure personal protection from chemical plant gases and corrosive liquids
- KU6.** the benefits and use of flame-proof equipment/enclosures
- KU7.** the operation of grinding equipment, digesters, precipitation and crystallization setups, filters, and mixer settlers in the chemical plant
- KU8.** the safety procedures applicable to Rare Earth Extraction Plant (REEP)
- KU9.** the processes, such as procurement, store management, inventory management, quality management, etc
- KU10.** the appropriate personnel to be contacted for query resolution

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the critical equipment observations and report them to the supervisor
- GS2.** write requisition to supervisor on the requirement of apparatus, tools etc
- GS3.** read and interpret blueprint, control diagrams and SOP documents
- GS4.** communicate politely and professionally
- GS5.** listen attentively to understand the information/ instructions being shared
- GS6.** plan and prioritize tasks to ensure their timely completion
- GS7.** systematize all process/equipment in 5s plan to process fast
- GS8.** identify potential work-related problems and take appropriate preventive measures
- GS9.** evaluate all possible solutions to a problem and implement the best one
- GS10.** make appropriate decisions concerning work within the applicable limits of authority
- GS11.** follow the organizational procedure during workplace emergencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Supply electricity to the installed system and equipment</i>	20	34	-	15
PC1. use the appropriate Personal Protective Equipment (PPE) including Gum boots, Polyurethane (PU) gloves, ear muffs and eye protection	3	5	-	2
PC2. follow the appropriate electrical safety precautions at the rare earth extraction plant	1	3	-	1
PC3. obtain the work permit from the authorized person following the applicable procedure	1	2	-	1
PC4. maintain the electrical checkpoints	1	-	-	-
PC5. ensure the relevant electrical parameters are aligned with the process requirement of 3-phase induction motors at various locations in the rare earth extraction plant	2	4	-	2
PC6. check the circuit breakers, isolators, bus bars, wiring, fuses, earthing, switchboard, control panels, relays, and diesel generators for the correct functioning	3	4	-	3
PC7. follow the applicable safety standards in the earthing system, armoured cables and flexible cables as per the Indian Electricity Rules	1	3	-	1
PC8. maintain the electrical equipment, such as motors, fans, lighting, ACs, heaters, compressors, MCC Panels, LDB, PDB, etc	3	5	-	2
PC9. inspect the electrical equipment to identify electrical risks, hazards, defects and the needs for the adjustment/repair	3	5	-	2
PC10. modify the parameters of electrical equipment, if required to ensure alignment with the prescribed standards	2	3	-	1
<i>Perform testing, maintenance and operations</i>	10	16	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. carry out maintenance of different types of motors and pumps, e.g. agitators motors, crystallize motors, centrifugal pumps, evaporator pumps, process water pumps, etc	3	4	-	1
PC12. carry out predictive, preventive and breakdown maintenance of diesel generators, motors, circuit breakers, isolators, bus bars, MCC panels, switchboards, wiring, protective relays, etc	3	4	-	2
PC13. carry out testing and identification of electrical systems for proper functioning of electrical wiring, equipment, and fixtures	2	4	-	1
PC14. repair or replace defective wiring in power sources, lighting, AC and safety control systems, following the standard procedure	1	3	-	1
PC15. follow appropriate safety precautions to ensure personal protection from corrosive liquids and gases found in the rare earth extraction plant	1	1	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3109
NOS Name	Test and operate electrical equipment and motors at the rare earth extraction plant
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N3110: Test and operate electrical equipment and motors at the main receiving sub-station

Description

This occupational standards unit is about testing and operating electrical equipment and motors at the main receiving sub-station

Scope

The scope covers the following :

- Manage the sub-station operations
- Maintain the sub-station
- Carry out documentation
- Monitor and inspect

Elements and Performance Criteria

Manage the sub-station operations

To be competent, the user/individual on the job must be able to:

- PC1.** check different equipment in the power system, e.g. transformers, circuit breakers, CT, PT, Surge Arrestors, Isolators, Relays etc., for the correct functioning
- PC2.** maintain the protective tags and establish safe clearance for the sub-station equipment, i.e. Lockout-Tagout (LOTO)
- PC3.** carry out testing and operate the circuit breaker, Capacitor bank, Transformer, Lightning arrester, Isolating switches, Insulators, and Relays
- PC4.** undertake switching operations of the outgoing feeder and in the specified plant as per the standard procedure
- PC5.** use the appropriate control equipment, such as On Load Tap Changer and circuit breakers to regulate the flow of electricity through sub-stations and overhead distribution lines to the plant
- PC6.** perform the di-electric test for the transformer oil
- PC7.** perform incidental or related duties such as filling distilled water in batteries, housekeeping and attending telephone calls from different plants and offices

Maintain the sub-station

To be competent, the user/individual on the job must be able to:

- PC8.** maintain the overhead lines as per the SOP
- PC9.** perform maintenance of transformers, circuit breakers, capacitor bank, panels, bus bars, relays, CT, PT, isolators, etc.
- PC10.** carry out testing of relays following the secondary injection method
- PC11.** monitor different equipment performance parameters, such as voltage, current flow, and temperature to ensure they are within safe operating ranges
- PC12.** carry out maintenance of the transformer's cooling fans and OLTC

Qualification Pack

PC13. conduct appropriate tests, e.g. IR Value, Ratio Test, winding resistance, No Load Test, Short Circuit test, to check and maintain different transformer parameters

PC14. use a wooden ladder in the maintenance of the electrical platform structure

Carry our documentation

To be competent, the user/individual on the job must be able to:

PC15. review the maintenance requests from end users and follow up to ensure that work is completed correctly

PC16. report outages and load data to plants

PC17. record the daily temperatures and oil levels in transformer stations

PC18. communicate, co-ordinate and transmit switching orders via telephone

PC19. maintain appropriate records for work activities, e.g. work orders, log book, repair schedules, and accident reports

Monitor and inspect

To be competent, the user/individual on the job must be able to:

PC20. perform periodical and regular inspections of the station, both inside the control room and outside the switch yard to ensure appropriate facility operation and guard railing

PC21. monitor and control the power sources A/C & D/C lightings, rectifiers, phase angle regulators, protection relays and generators

PC22. monitor the current and potential type transformer to ensure its proper operation

PC23. inspect, operate and maintain sub-stations and distribution apparatus as per the applicable safety regulations

PC24. check earth pit resistance at the appropriate frequency

PC25. operate and maintain the diesel generator

PC26. follow appropriate safety precautions while working around high voltage electrical equipment, distribution panel boards, measuring and indicating instruments

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the importance of following the daily maintenance checklist

KU2. the consequences of not following defined procedures/work instructions

KU3. the applicable escalation matrix and work reporting procedures

KU4. the importance of maintaining the equipment and the loss to the organization due to equipment damage or breakdown

KU5. the implications of delays in the mining processes

KU6. the Directorate General of Mines Safety (DGMS)'s safety guidelines specific to electricity-relevant operations

KU7. the basic principles of electrical energy

KU8. the Operations and Maintenance (O&M) of electrical systems

KU9. the use of different types of hand tools and electrical equipment

KU10. the appropriate safety measures to be taken for different electrical hazards

KU11. the statutory requirements for electrical systems

Qualification Pack

- KU12.** the causes and prevention of electrical defects/ malfunctions
- KU13.** the working of electrical systems and machines
- KU14.** the functioning and uses of Diesel Generator (DG), mobile lighting equipment, high mast, etc
- KU15.** the installation and handling of safety devices
- KU16.** the uses of the Programmable Logic Controller (PLC)
- KU17.** the standards and process of illumination, noise and vibration survey
- KU18.** the interlocking of equipment as per the Central Electricity Authority regulations
- KU19.** the appropriate equipment used in sub-station construction and maintenance activities
- KU20.** the importance of using gravel to pave the sub-station surfaces

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make note of work-related observations
- GS2.** interpret the relevant drawings, sketches, symbols, measurements and Original Equipment Manufacturer (OEM) manuals
- GS3.** use the online Enterprise Resource Planning (ERP) systems under the guidance of the supervisor
- GS4.** communicate politely and professionally
- GS5.** use effective questioning techniques to extract maximum information about different tasks
- GS6.** plan daily tasks appropriately to ensure their timely completion
- GS7.** undertake different activities as per the applicable task lists and schedules with minimum supervision
- GS8.** make appropriate decisions concerning the area of work
- GS9.** follow the 5s principles at work
- GS10.** use reasoning skills to identify and resolve basic problems at work
- GS11.** assist in identifying and implementing appropriate solutions to work-related problems
- GS12.** follow the organizational protocol during emergencies

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage the sub-station operations</i>	8	15	-	6
PC1. check different equipment in the power system, e.g. transformers, circuit breakers, CT, PT, Surge Arrestors, Isolators, Relays etc., for the correct functioning	1	3	-	1
PC2. maintain the protective tags and establish safe clearance for the sub-station equipment, i.e. Lockout-Tagout (LOTO)	2	2	-	1
PC3. carry out testing and operate the circuit breaker, Capacitor bank, Transformer, Lightning arrester, Isolating switches, Insulators, and Relays	1	3	-	1
PC4. undertake switching operations of the outgoing feeder and in the specified plant as per the standard procedure	1	2	-	1
PC5. use the appropriate control equipment, such as On Load Tap Changer and circuit breakers to regulate the flow of electricity through sub-stations and overhead distribution lines to the plant	1	2	-	1
PC6. perform the di-electric test for the transformer oil	1	2	-	1
PC7. perform incidental or related duties such as filling distilled water in batteries, housekeeping and attending telephone calls from different plants and offices	1	1	-	-
<i>Maintain the sub-station</i>	7	15	-	6
PC8. maintain the overhead lines as per the SOP	1	2	-	-
PC9. perform maintenance of transformers, circuit breakers, capacitor bank, panels, bus bars, relays, CT, PT, isolators, etc.	1	2	-	1
PC10. carry out testing of relays following the secondary injection method	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. monitor different equipment performance parameters, such as voltage, current flow, and temperature to ensure they are within safe operating ranges	1	2	-	1
PC12. carry out maintenance of the transformer's cooling fans and OLTC	1	2	-	1
PC13. conduct appropriate tests, e.g. IR Value, Ratio Test, winding resistance, No Load Test, Short Circuit test, to check and maintain different transformer parameters	1	4	-	1
PC14. use a wooden ladder in the maintenance of the electrical platform structure	1	1	-	1
<i>Carry our documentation</i>	4	5	-	3
PC15. review the maintenance requests from end users and follow up to ensure that work is completed correctly	1	1	-	1
PC16. report outages and load data to plants	1	1	-	1
PC17. record the daily temperatures and oil levels in transformer stations	-	1	-	1
PC18. communicate, co-ordinate and transmit switching orders via telephone	-	1	-	-
PC19. maintain appropriate records for work activities, e.g. work orders, log book, repair schedules, and accident reports	2	1	-	-
<i>Monitor and inspect</i>	11	15	-	5
PC20. perform periodical and regular inspections of the station, both inside the control room and outside the switch yard to ensure appropriate facility operation and guard railing	2	3	-	1
PC21. monitor and control the power sources A/C & D/C lightings, rectifiers, phase angle regulators, protection relays and generators	2	3	-	1
PC22. monitor the current and potential type transformer to ensure its proper operation	1	1	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. inspect, operate and maintain sub-stations and distribution apparatus as per the applicable safety regulations	3	2	-	1
PC24. check earth pit resistance at the appropriate frequency	1	2	-	1
PC25. operate and maintain the diesel generator	1	2	-	-
PC26. follow appropriate safety precautions while working around high voltage electrical equipment, distribution panel boards, measuring and indicating instruments	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3110
NOS Name	Test and operate electrical equipment and motors at the main receiving sub-station
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1810: Follow the applicable health, safety and environmental guidelines (Placer Mining)

Description

This unit is about following the applicable health, safety and environmental guidelines.

Scope

The scope covers the following :

- Maintain personal health and safety
- Assist in hazard management
- Work on equipment and machinery
- Check the first aid box, firefighting and safety equipment
- Report incidents

Elements and Performance Criteria

Maintain personal health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers
- PC2.** ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments
- PC3.** clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes
- PC4.** follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC5.** follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking
- PC6.** use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose
- PC7.** identify six directional hazards at the workplace and take decisions accordingly
- PC8.** follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas
- PC9.** undertake the Take-5 (Personal Risk Assessment) before the commencement of any work
- PC10.** lift objects carefully maintaining the correct body posture to avoid injury
- PC11.** use self-rescue apparatus appropriately when required
- PC12.** work in the assigned work zone in the designated work area
- PC13.** follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc
- PC14.** wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant
- PC15.** use caution signs as applicable and obey safety signs, stickers, and tags

Qualification Pack

PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact

PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help

Assist in hazard management

To be competent, the user/individual on the job must be able to:

PC18. label and store chemicals appropriately

PC19. display the MSDS for the hazardous chemicals in storage and handling areas

PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines

PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre

PC22. maintain regular and open communication with co-workers and the employer

Work on equipment and machinery

To be competent, the user/individual on the job must be able to:

PC23. check and ensure every instrument, apparatus and equipment is AERB approved

PC24. operate the appropriate tools, equipment and machinery with proper training and authorization

Check the first aid box, firefighting, and safety equipment

To be competent, the user/individual on the job must be able to:

PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible

PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas

PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors

PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite

Report incidents

To be competent, the user/individual on the job must be able to:

PC29. identify and report potential hazards/ unsafe conditions and practices

PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.

PC31. report the emergencies and any safety incidents as per the organizational procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of chemical plants and their operations

KU2. the importance of time-keeping, discipline and punctuality

KU3. various chemical processes carried out in a rare earth extraction plant

KU4. the impact of violating applicable safety procedures

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- KU5.** the relevant quality norms
- KU6.** different types of individual rare earth chemical processes and the use of associated equipment
- KU7.** different types of effluent treatment processes and the use of associated equipment
- KU8.** the use of different water purifying and waste treatment agents
- KU9.** the hazards and the appropriate safety measures in acid/ alkali handling activities and usage of relevant PPE
- KU10.** hazards and safety aspects involved in acid/ alkali handling activities and usage of relevant PPE
- KU11.** the internal safety organization and the role of the safety committee, workforce inspector and AERB, etc
- KU12.** the importance of sensitization towards different genders and persons with disabilities (PWD)
- KU13.** the Importance of following infection control policies, '5-S' practices, and waste management as prescribed by regulatory authorities like the Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.
- KU14.** the importance of water/material/energy conservation and management
- KU15.** the importance of Social Management Plan (SMP) and Environment Management Plan (EMP) prepared by the organization, as per DGMS guideline
- KU16.** the importance of wearing own TLD or pocket dosimeter (to be obtained from the Health Physics Unit) in the active area
- KU17.** the importance of staying alert and attentive to the work surroundings
- KU18.** the importance of staying in the assigned work zone in the designated work area
- KU19.** the benefits of making the safety rules part of new employee orientation
- KU20.** the importance of contacting the supervisor or manager for instruction, guidance, or training when required

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal health and safety</i>	16	24	-	11
PC1. follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers	1	2	-	1
PC2. ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments	1	1	-	1
PC3. clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes	1	1	-	-
PC4. follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	-
PC5. follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking	1	2	-	-
PC6. use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose	1	2	-	-
PC7. identify six directional hazards at the workplace and take decisions accordingly	1	2	-	1
PC8. follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas	1	2	-	1
PC9. undertake the Take-5 (Personal Risk Assessment) before the commencement of any work	1	2	-	1
PC10. lift objects carefully maintaining the correct body posture to avoid injury	-	1	-	-
PC11. use self-rescue apparatus appropriately when required	1	1	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. work in the assigned work zone in the designated work area	1	1	-	1
PC13. follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc	1	1	-	1
PC14. wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant	1	1	-	1
PC15. use caution signs as applicable and obey safety signs, stickers, and tags	1	1	-	-
PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact	1	1	-	1
PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help	1	1	-	1
<i>Assist in hazard management</i>	5	8	-	4
PC18. label and store chemicals appropriately	1	1	-	1
PC19. display the MSDS for the hazardous chemicals in storage and handling areas	1	2	-	1
PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines	1	1	-	-
PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre	1	2	-	1
PC22. maintain regular and open communication with co-workers and the employer	1	2	-	1
<i>Work on equipment and machinery</i>	2	4	-	1
PC23. check and ensure every instrument, apparatus and equipment is AERB approved	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. operate the appropriate tools, equipment and machinery with proper training and authorization	1	2	-	1
<i>Check the first aid box, firefighting, and safety equipment</i>	4	10	-	2
PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible	1	3	-	-
PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas	1	2	-	-
PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors	1	2	-	1
PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite	1	3	-	1
<i>Report incidents</i>	3	4	-	2
PC29. identify and report potential hazards/ unsafe conditions and practices	1	2	-	-
PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.	1	1	-	1
PC31. report the emergencies and any safety incidents as per the organizational procedure	1	1	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1810
NOS Name	Follow the applicable health, safety and environmental guidelines (Placer Mining)
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1811: Work collaboratively and effectively in a team

Description

This unit is about the importance of teamwork in the workplace.

Scope

The scope covers the following :

- Work effectively with co-workers
- Communicate effectively with co-workers
- Practice inclusiveness at work

Elements and Performance Criteria

Work effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC1.** identify own direct and indirect responsibilities, e.g. duties concerning team members
- PC2.** undertake appropriate activities to promote cooperation and good relationships with co-workers
- PC3.** mentor and assist subordinates in the execution of their work responsibilities
- PC4.** identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures
- PC5.** follow the recommended practices to avoid and resolve conflicts at work
- PC6.** interact effectively and cooperate with co-workers
- PC7.** manage workplace conflicts, ensuring no impact on work
- PC8.** identify and take responsibility for own mistakes
- PC9.** follow the organizational policies and procedures
- PC10.** follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.
- PC11.** exchange feedback with peers and other team members to improve the quality of work

Communicate effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC12.** follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel
- PC13.** communicate clearly and politely to ensure effective communication with co-workers
- PC14.** practice active listening to understand the conveyed information and instructions
- PC15.** practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication
- PC16.** maintain the professional standard of communication, e.g. avoiding the use of offensive language at work

Practice inclusiveness at work

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC17.** empathize with Persons with Disabilities (PwD)
- PC18.** practice gender equality at work
- PC19.** maintain respectful behaviour and etiquette at work
- PC20.** listen to and acknowledge the feelings, concerns, opinions, and ideas of others

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of teamwork and the consequence of poor teamwork on project outcomes, timelines, safety, etc
- KU2.** the duties and rights of workers
- KU3.** the importance of creating a healthy and cooperative work environment
- KU4.** the importance of proper and effective communication and the adverse effects of poor communication on quality, timeliness, safety, etc
- KU5.** different modes of communication used at work
- KU6.** the importance and need of supporting co-workers facing problems for the smooth functioning of work
- KU7.** the fundamental concept of gender equality
- KU8.** how to recognize and be sensitive to issues of disability, culture and gender
- KU9.** how to show empathy to PwD appropriately
- KU10.** legislation, policies, and procedures relating to gender sensitivity and cultural diversity including their impact on the area of operation
- KU11.** the appropriate behaviour to be practised with coworkers
- KU12.** the appropriate communication skills and leadership qualities
- KU13.** how to be open to listening to and understanding the other person's point of view

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with co-workers</i>	15	30	-	10
PC1. identify own direct and indirect responsibilities, e.g. duties concerning team members	3	5	-	3
PC2. undertake appropriate activities to promote cooperation and good relationships with co-workers	1	2	-	1
PC3. mentor and assist subordinates in the execution of their work responsibilities	1	2	-	1
PC4. identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures	3	4	-	2
PC5. follow the recommended practices to avoid and resolve conflicts at work	1	2	-	1
PC6. interact effectively and cooperate with co-workers	1	1	-	-
PC7. manage workplace conflicts, ensuring no impact on work	1	1	-	-
PC8. identify and take responsibility for own mistakes	1	3	-	1
PC9. follow the organizational policies and procedures	1	3	-	-
PC10. follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.	1	4	-	1
PC11. exchange feedback with peers and other team members to improve the quality of work	1	3	-	-
<i>Communicate effectively with co-workers</i>	9	10	-	5
PC12. follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel	2	3	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. communicate clearly and politely to ensure effective communication with co-workers	1	2	-	-
PC14. practice active listening to understand the conveyed information and instructions	2	1	-	1
PC15. practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication	2	2	-	1
PC16. maintain the professional standard of communication, e.g. avoiding the use of offensive language at work	2	2	-	1
<i>Practice inclusiveness at work</i>	6	10	-	5
PC17. empathize with Persons with Disabilities (PwD)	2	2	-	2
PC18. practice gender equality at work	2	2	-	2
PC19. maintain respectful behaviour and etiquette at work	1	3	-	1
PC20. listen to and acknowledge the feelings, concerns, opinions, and ideas of others	1	3	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1811
NOS Name	Work collaboratively and effectively in a team
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3107.Test and operate electrical equipment and motors at the dredge and supplementary mining plant	30	50	0	20	100	20
MIN/N3108.Test and operate electrical equipment and motors at the mineral separation plant	30	50	0	20	100	20
MIN/N3109.Test and operate electrical equipment and motors at the rare earth extraction plant	30	50	0	20	100	15
MIN/N3110.Test and operate electrical equipment and motors at the main receiving sub-station	30	50	0	20	100	15
MIN/N1810.Follow the applicable health, safety and environmental guidelines (Placer Mining)	30	50	0	20	100	10
MIN/N1811.Work collaboratively and effectively in a team	30	50	0	20	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	200	330	-	120	650	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.